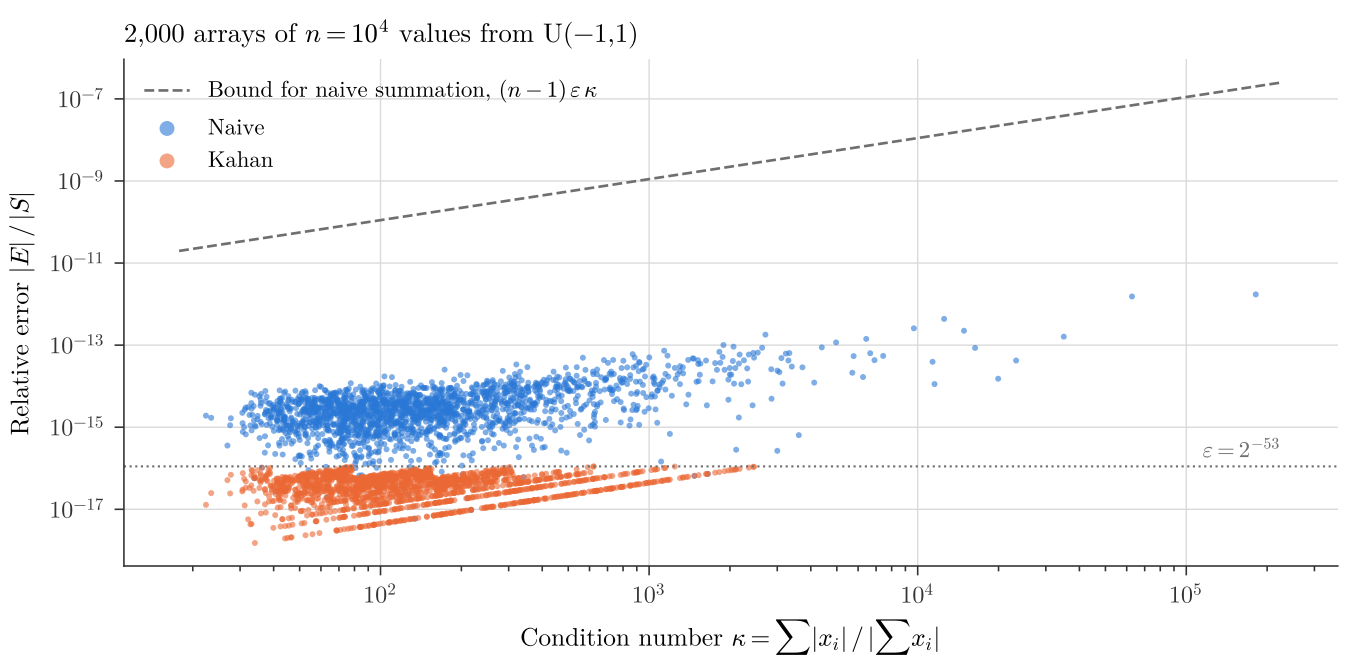

2,000 arrays of $n = 10^4$ values from U(−1,1)
Bound for naive summation, $(n-1)\,\varepsilon\,\kappa$
Naive
Kahan
$\varepsilon = 2^{-53}$
Relative error $|E| / |S|$
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{-17}
10^{2}
10^{3}
10^{4}
10^{5}
Condition number $\kappa = \sum|x_i| / |\sum x_i|$